Physicochemical EDA integrated with Structure-Selectivity Relationships (SSR)

AKR1C3 versus AKR1C2 | Exploratory computational medicinal chemistry analysis | 27 October 2026

Study design

The same 416 curated, stereochemistry-aware structures previously included in the Kennard-Stone benchmark were analyzed. The response was delta pIC50 = log10(IC50[AKR1C2]/IC50[AKR1C3]); a positive value indicates preferential inhibition of AKR1C3. RDKit descriptors, Bemis-Murcko scaffolds, 2048-bit chiral Morgan fingerprints and one-cut matched molecular-pair analysis were used. Original measurements and training datasets were not changed.

Summary of computational findings

Measurement / analysis	Result
Compound-level structures; structural scaffolds	416; 93
High-similarity pairs at Tanimoto >= 0.85	41
Same-DOI close-pair cliffs (>=10-fold; Tanimoto >=0.85)	2
One-cut matched molecular pairs from same DOI	707 pairwise comparisons
Same-DOI MMP comparisons with >=10-fold selectivity difference	87 non-independent comparisons
Enantiomer examples supported by same DOI	3 specified stereo-to-stereo comparisons

Chemical interpretation

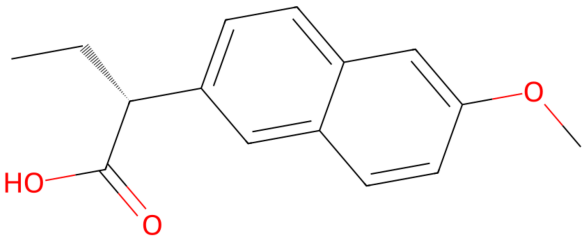
Stereochemical control: CHEMBL4435662 (R) and CHEMBL4466610 (S) represent enantiomers of a methoxynaphthylbutanoic-acid analogue. The R isomer has AKR1C3 IC50 = 110 nM and AKR1C2 IC50 = 48,100 nM, versus 120 nM and 1,720 nM for S. Selectivity differs by 30.5-fold, driven predominantly by sparing of AKR1C2 (approximately 28-fold weaker AKR1C2 inhibition in R). This observation cannot be captured by molecular weight or conventional Rule-of-Five descriptors, which are identical for the two enantiomers. Source DOI: 10.1021/acs.jmedchem.6b00160.

Substituent-chemistry example: Replacement of a methylsulfonyl group with a trifluoromethylsulfonyl group in a matched indomethacin-related pair (CHEMBL2323489 to CHEMBL2323490) coincides with a 171.7-fold higher selectivity ratio, from AKR1C3 IC50 12,000 to 210 nM and AKR1C2 IC50 16,680 to 50,130 nM. Source DOI: 10.1021/jm3017656. This is observational chemical SAR, not a demonstrated binding mechanism.

Descriptor confounding: Across all 416 compounds, cLogP versus selectivity has Spearman rho = 0.324. Within the largest scaffold (111 compounds), the correlation is only rho = 0.109. Global correlations therefore reflect chemical-series composition as well as possible within-series effects.

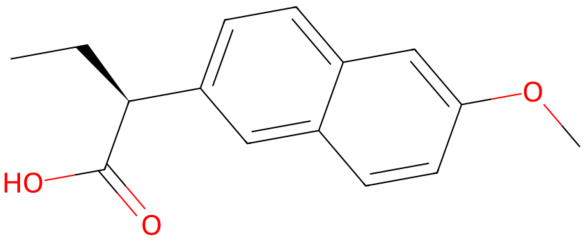
Representative chemically defined structure pairs

CHEMBL4435662 (R)



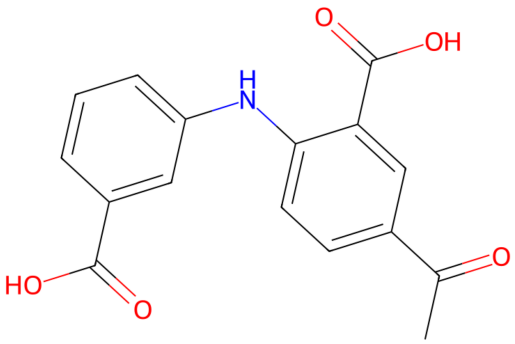
C3: 110 nM C2: 48100 nM
Delta pIC50: 2.64

CHEMBL4466610 (S)



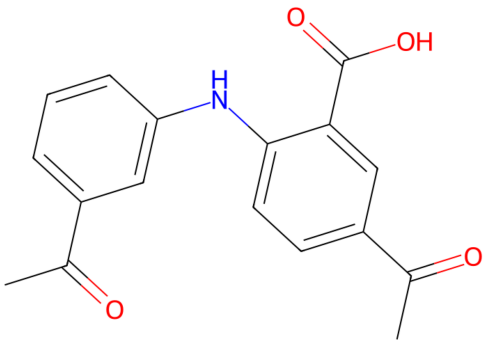
C3: 120 nM C2: 1720 nM
Delta pIC50: 1.16

CHEMBL2041554



C3: 40 nM C2: 6410 nM
Delta pIC50: 2.20

CHEMBL2040993



C3: 650 nM C2: 1880 nM
Delta pIC50: 0.46

Figure 1. RDKit-generated chemical structures with original compound IDs and curated paired IC50 values. Upper: enantiomer contrast; lower: substantial aryl carboxyl/acetyl substitution-pattern differences. The lower comparison is not a single-position isomerization.

Physicochemical association depends on scaffold context

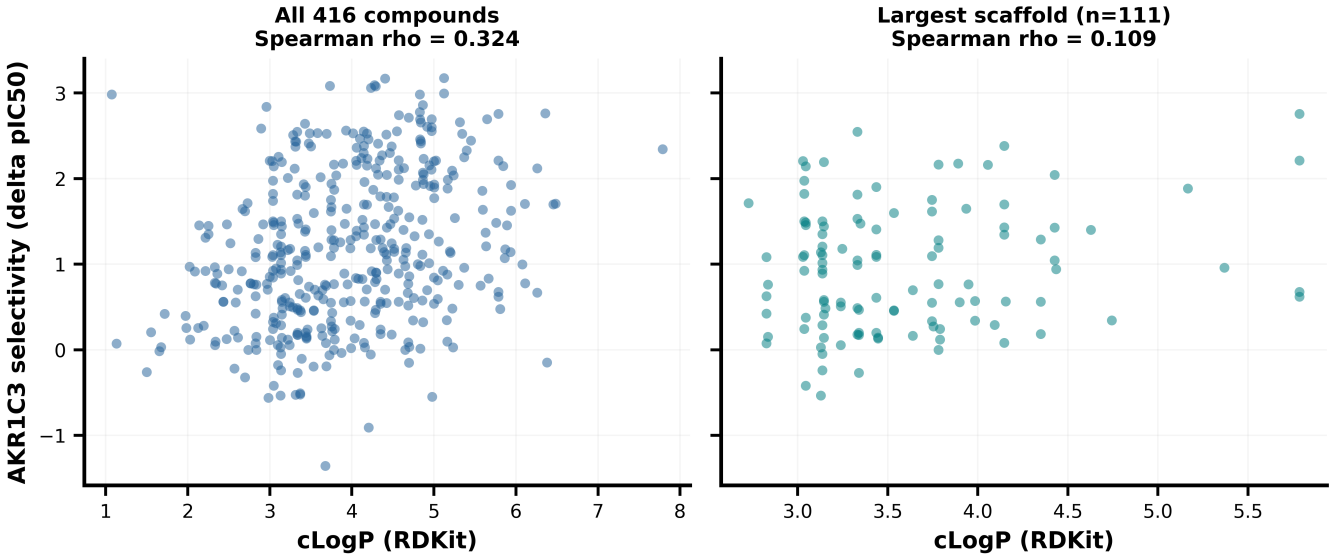


Figure 2. Compound-level cLogP and experimental selectivity in the full dataset and largest Bemis-Murcko series. No causal effect is inferred.

Chemical neighborhood selectivity cliffs

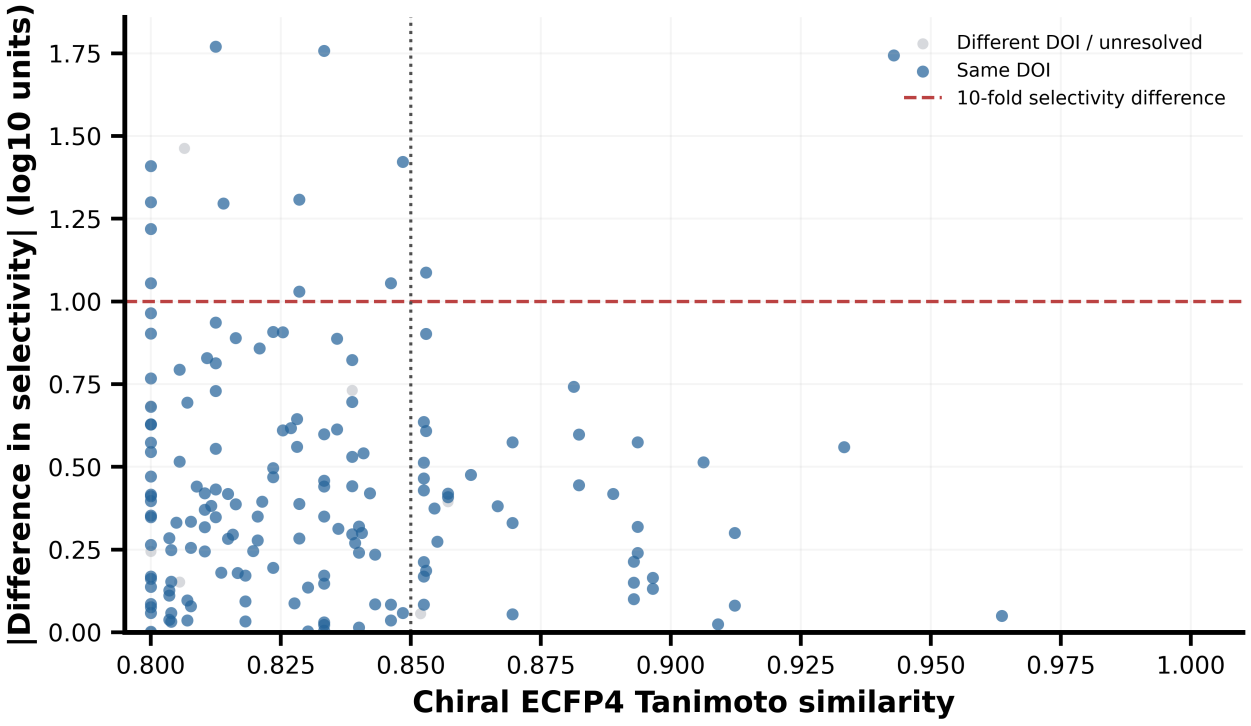


Figure 3. 168 molecular pairs have chiral ECFP4 Tanimoto >= 0.80; 41 have Tanimoto >= 0.85. Two pairs also differ by at least 10-fold in selectivity and have shared publication DOI. A common DOI is a partial provenance check, not proof of identical assays.

Stereochemistry and isoform-specific inhibition

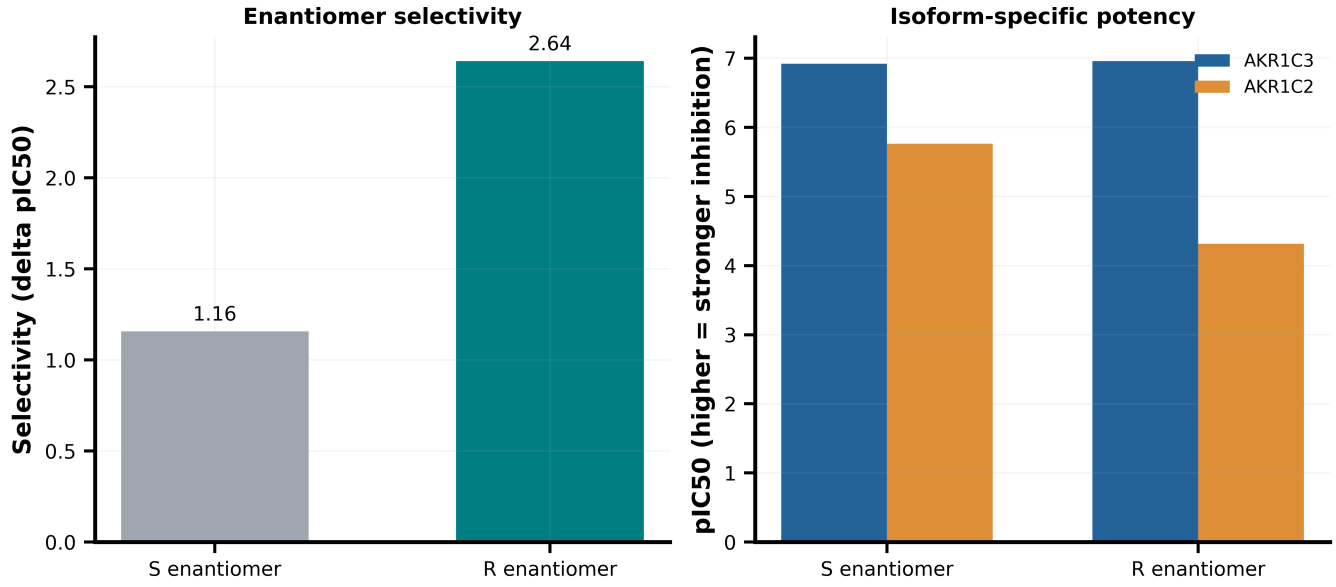


Figure 4. The R/S difference is dominated by the AKR1C2 counter-activity rather than a major shift in AKR1C3 inhibition.

Scaffold heterogeneity

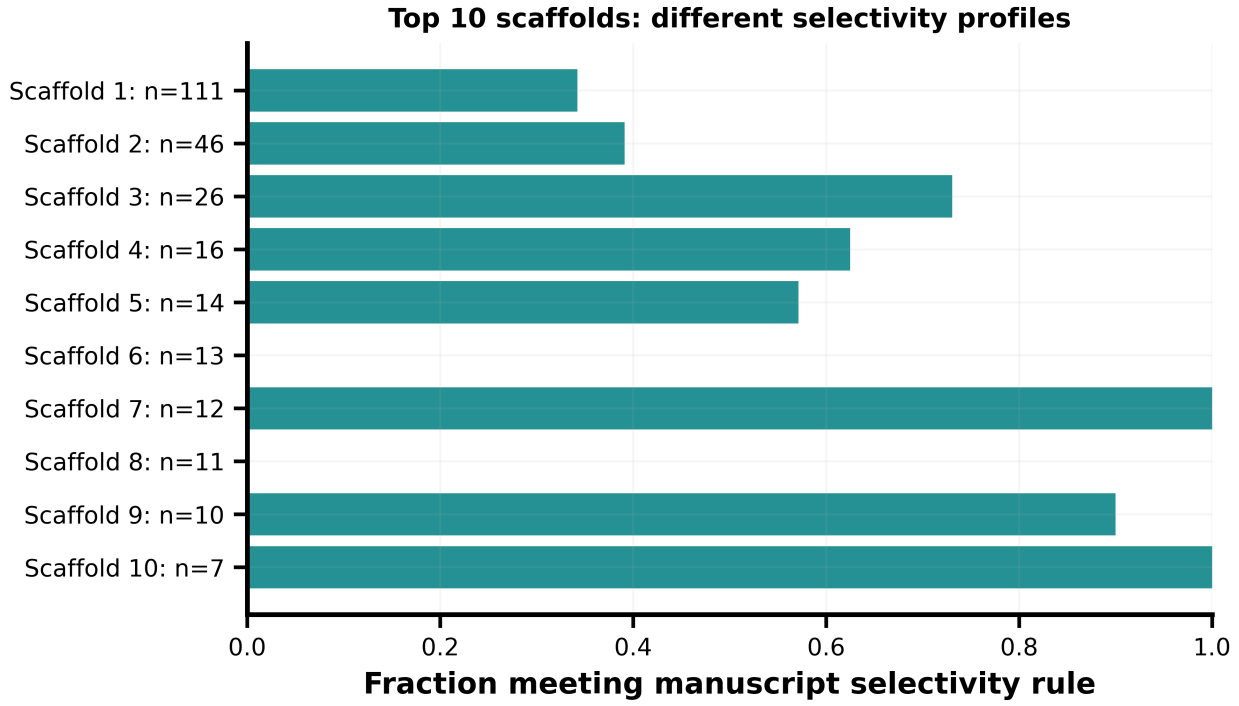


Figure 5. Different core series occupy different experimental selectivity regimes. Rules learned from a pooled dataset are not automatically transferable between scaffolds.

Interpretive conclusions for MSc Chemistry

1. Physicochemical descriptors map the broader chemical space, but do not individually establish molecular mechanisms governing isoform selectivity.

2. Steric arrangement, functional-group identity and substitution patterns can create substantial selectivity changes among structurally related inhibitors.

3. Experimental AKR1C3 and AKR1C2 potency changes must be considered separately; a selectivity improvement may result from selective loss of AKR1C2 inhibitory activity.

4. Selectivity cliffs and matched pairs yield chemically testable hypotheses. Their interpretation requires checking assay substrates, measurement protocols, and replicate consistency in each original source.

Data provenance and limitations

The database remains observational. Compound-level IC50 values were aggregated to medians before SSR comparison. For two structures with multiple measurements, median selectivity log-ratios and ratios of separately aggregated IC50 medians differ slightly (maximum 0.063 log units). All original source and DOI identifiers are retained in accompanying CSV outputs. One-cut matched-pair comparisons are non-independent and are not causal tests or counts of unique discoveries. Stated stereoisomer comparisons distinguish fully specified R/S pairs from racemate or unspecified configuration.

Source evidence for selected examples

https://doi.org/10.1021/acs.jmedchem.6b00160

https://doi.org/10.1021/jm201547v

https://doi.org/10.1021/jm3017656

https://doi.org/10.1021/acs.jmedchem.3c00732

Files and reproducibility

REPRODUCE_SSR.py contains all RDKit fingerprint, matched molecular-pair, stereochemical and descriptive statistical calculations. CSV files provide complete SMILES, assay endpoints, publication DOI, pair similarities, transformation SMILES, and selectivity fold changes. Generated figures are supplied separately as high-resolution PNG and vector PDF.